Diversity experiments

In the early 1970s, Connell and American ecologist Daniel Janzen published an explanation of the degree of tree diversity in tropical forests: the Janzen–Connell hypothesis. Connell mapped trees in two rain forests in North Queensland, Australia, and found that seedlings tended to be less successful when their nearest neighbor was of the same species. Each species is targeted by specific herbivores and pathogens, which will also eat or attack smaller, weaker individuals of the species nearby. This prevents "clumping" of one tree species.

In 1978, Connell proposed the intermediate disturbance hypothesis (IDH). This states that both high and low levels of disturbance reduce species diversity in an ecosystem, so the greatest range of species can be expected between those extremes. Several studies support IDH. One, carried out in waters off Western Australia, examined the effects of wave disturbance on diversity. Species diversity was found to be low both at exposed offshore sites and at sheltered sites.

See also: Ecological niches • A modern view of diversity • Animal behavior • The ecosystem • Niche construction